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Reading Assignment

Ordinary differential equations are a representation of how variables change with respect to another variable. ODE applications are often used to describe systems such as heat transfer, fluid flow, electrical circuits, and more. ODEs are very common in scientific fields, especially engineering. Due to the complexity of these equations, it is extremely difficult and tedious to solve them by hand. Oftentimes, it is much more efficient to have a program like MATLAB compute the solutions.

When looking toward problem-solving software like MATLAB, there are a few different routes you can take when computing these values. The main avenues separate into ODE23, ODE45, and ODE15 methods. These methods are built into MATLAB and allow you to solve for solutions easily, but all three operate in slightly different ways.

The ODE23 method is an adaptive Runge-Kutta solver that evaluates the derivative you are analyzing over numerous points and uses this information to easily estimate the solutions while accounting for any errors. What makes this method unique and so effective is the automatic adjustment of step intervals. Some methods require you to adjust the size between each analysis manually. This may leave too much room between points and potentially lead to a poor estimate. The ODE23 method does this automatically. MATLAB

tracks the rate at which the function is changing. When the plot is relatively smooth and linear, it takes fewer steps; when it varies rapidly, it takes smaller steps for a better estimate. This method allows the program to solve the equation quickly and effectively.

The ODE45 method is quite similar to the ODE23 method in the way it works. However, there is a key difference that allows it to compute values with a higher degree of accuracy. The 4 and 5 in ODE45 refer to the fourth- and fifth-order solutions of the equation. How these solutions are used is quite simple. MATLAB computes both, then compares them and estimates the error. If the estimated error is small, it will take a larger next step. If the estimated error is large, it takes a smaller step. This is the same process as the ODE23 method, but instead of comparing the second- and third-order solutions, it is comparing the fourth- and fifth-order solutions, allowing for a higher degree of certainty.

The last method we will analyze is ODE15, which differs from ODE23 in a way similar to how ODE23 differs from ODE45. What makes ODE15 unique is that it solves the equations using implicit methods rather than estimations like the two previous examples. Basically, ODE15 solves for each value along the entire solution. This has advantages and disadvantages. The downside is that it is much more complicated and can take much longer to compute the values. However, it is often very stable and can perform well even when computing very long operations.

The Runge-Kutta method is integral to the performance of ODE23 and ODE45. Runge-Kutta methods work by checking the slope at multiple points during each step to estimate the average slope. This makes them effective because they constantly check and adjust the step size in ODE23 and ODE45.

